

ALIFE - Practical work

Delivery Deadline: June 14, 2026.**Support:** Report in digital format (PDF or DOC), source code and executable file(s)

The medium size project can be done optionally by each group of students (max 4 elements)
For each work, the source code, an executable and a mini report (up to 15 pages) explaining the work performed must be delivered.

Prev Modeling

The goal is to create an artificial life simulation environment, with virtual beings called **Zizoids** that can survive in their environment, and their predators called **Wsiloids** that are found in the same environment.

Zizoids, need brains capable of processing input from the outside world around them. This should be implemented using an artificial neuronal network.

Zizoids must be able to recognize food when they find it to move in the right direction and at the same time escape their predators.

The **Zizoids** have eyes, which provide input to the ANN (artificial neural network), which forms their brain and which is directly attached to their legs. As output is expected the following:

Output	Result
Move left leg	Moves front right
Move right leg	Moves front left
Move both legs	Moves forward
Move both legs backwards	Backward - Moves backward
Move right leg laterally	Rotate 90%
Do not move	Does not move

ANN receives input from the **Zizoids** vision system and stomach (hunger) and this information will be used to produce outputs transmitted to the legs. So, the ANN receives input from the **Zizoid's** vision system and produces, depending on its configuration (morphological design and synaptic values), an output in the form of signals transmitted to the legs.

The number of ANN neurons depends on the type of network and at least 9 neurons will be used for input and 6 for output.

Each input must be unique: Food or a front-left **Wsiloids** must produce a different signature than a front-right **Wsiloids**.

For simplicity, we consider that each cell in the virtual world can contain only one item at any given time.

Each **Zizoid** will be able to see the 8 cells in its neighborhood, regardless of the direction in which it is moving. Examples:

			OO	OO	OO
			→Z	OO	OO
OO	OO	OO	OO	OO	OO
OO	OO	OO			
OO	↑ Z	OO			

Being :

- 'Arrow' - direction of movement
- Z - **Zizoid**
- OO - cells in the visual range

In each cycle (generation), each **Zizoid** has to: look at the space ahead of it; think about what you saw; perform an action. In each generation the energy value of each schizoid is decreased by 1 unit. When this parameter reaches the value 0, the **Zizoid** dies and is removed from the simulation.

The mechanism to be used for an increasingly accurate and capable breed of **Zizoid** is natural selection. When a **Zizoid** moves to a cell where another **Zizoid** is located, a random number is generated. Given the probability pre-set by the user before the start of the simulation, the system decides whether or not to reproduce. The **Zizoid** are hermaphrodites.

When reproduction takes place, a new **Zizoid** whose genetic code is derived from its parents will be placed in any free space in the environment. These will remain in the cells they are in.

The crossover should be done in accordance with a criterion that must be clearly justified. Part of the genome will come from the parent **Zizoid**; another part will come from the mother **Zizoid**. During the process of copying the parent **Zizoid** genes, mutation may occur at the established rate. The number of alleles will depend on the number of neurons in the network.

Wsiloids simply move without any order through the defined environment. They have no brain or DNA and never die. When a **Wsiloid** decides to move to a given cell in a certain direction, if the new position is occupied by another **Wsiloid** then it will remain in its current position. If the new position is occupied by a **Zizoid** then the **Zizoid** will die.

Required System Features:

Mandatory configurable parameters:

- Initial energy of each new individual
- Mutation rate
- Likelihood of reproduction occurring
- Initial number of **Zizoids**
- No. of **Wsiloids**
- Amount of energy gained from consuming an **Wsiloids**
- Amount of energy spent on reproduction
- Maximum number of **Zizoids**.

There must be a possibility to evaluate the genome of the **Zizoids** more successfully as well as the middle **Zizoids** and the worst **Zizoids**. The function of assessing the adequacy of the **Zizoids** is equal to measuring its age: the older it is, the more appropriate it will be.

The simulation is to enable a world that has a **minimum size of 300 cells**.